

Monkey See, Monkey Diffuse: Alternative Architectures for fMRI Image Reconstruction

George Liu (georgeli) Ryan Ding (ryanding) Kyle Lee (kylel)

March 2026

1 Introduction

Decoding visual stimuli from fMRI signals is a challenging problem at intersection of neuroscience and generative modeling. State-of-the-art approaches utilizing diffusion still achieve limited success with classification accuracy $< 28\%$ (Chen et al)[1]. We hypothesize that the backbone of diffusion models is particularly important in this setting with noisy inputs. To investigate this, we replace the conventional U-Net based architecture with the DiT, U-Vit, and PixArt- α as alternative backbones. Through these experiments we evaluate the performance of transformers, skip connections, and the importance of image realism in the context of fMRI image reconstruction.

2 Dataset, Task

Dataset(s):

- **Pretraining:** Similar to Chen et al.[1] we pretrain on Generic Object Decoding Dataset (GOD) which will provide us with 136,000 fMRI scan to image pairs.
- **Validation:** We validate on Brain, Object, Landscape Dataset (BOLD5000) which contains 5254 fMRI scan to image pairs and a holdout set from GOD.

Both of these datasets are publicly available and processed here as they were used in the Chen paper.[1]

Task: We replaced the U-Net backbone of the Chen et al. pipeline with the following alternatives: DiT, U-Vit, and PixArt- α . In the methods section we discuss why we chose these backbones. For each of these alternate architectures we conduct ablation studies on crop ratio and DDIM steps.

Metric(s): Measuring reconstruction quality is a challenging and subjective task. Therefore, to be robust we evaluate with 4 metrics:

- **50-way top-1 accuracy:** Similar to Chen et al and Ozcelik et al. we utilize a pre-trained ImageNet1K classifier to measure the 50-way top-1 accuracy, which measures out 50 possible classes how often the reconstructed image has the same highest probability class as the ground truth. We include this metric as it is the standard to benchmark against in this domain[2][1].
- **FID:** Frechet Inception captures the similarity of generated

images to real ones. We use this as our measure of "image quality" since it's one of the current state-of-the-art standards for measuring the quality of generative models.[4]

- **SSIM:** Structural Similarity Index measures the luminance, contrast, and structure (are patterns, edges, and textures aligned) between two images.[5] This is an appropriate method for judging reconstruction quality because generative models are not be able to reconstruct the exact same image but rather we want to judge if the general structure is similar.
- **PSM:** Perceptual Similarity Metric measures how often a reconstruction is perceptually closer to its correct ground truth than to other images in the dataset. The underlying distance is LPIPS (Learned Perceptual Image Patch Similarity), which compares images at the feature level using a pre-trained deep network (AlexNet) to approximate human perceptual judgments[6]. Higher PSM values indicate better perceptual fidelity.

We built off the mind-vis GitHub repository, which is the current state-of-the-art diffusion pipeline for fMRI image reconstruction.

3 Related Work

Generative Models in fMRI Image Reconstruction: Chen et al.[1] in 2023 designed the current state of the art for generating images from human fMRI data. They proposed a Sparse-Coded Masked Brain Modeling pipeline, which is a brain feature learner for vision decoding, with an LDM utilizing a U-Net backbone and achieved 50-way top-1 accuracy of 27.4%. We base our approach primarily on their pipeline. An earlier work in this space is from Ozcelik et al. 2022 which proposed a method for reconstructing images from fMRI signals using an Instance-Conditioned GAN (IC-GAN). Unlike prior approaches that typically capture either low-level visual details or high-level semantic content, their method jointly models both by decoding multiple latent variables (instance features, noise, and dense vectors) from brain activity via ridge regression.[2] We similarly do joint modeling. Belyi & Gaziv et al pioneered a method that enables unsupervised learning on unlabeled fMRI data by training both an encoder and decoder network in order to account for the sparsity of labeled fMRI data.[11] Later approaches found this to be unnecessary so we do supervised learning. One of the primary labeled datasets is BOLD5000 which

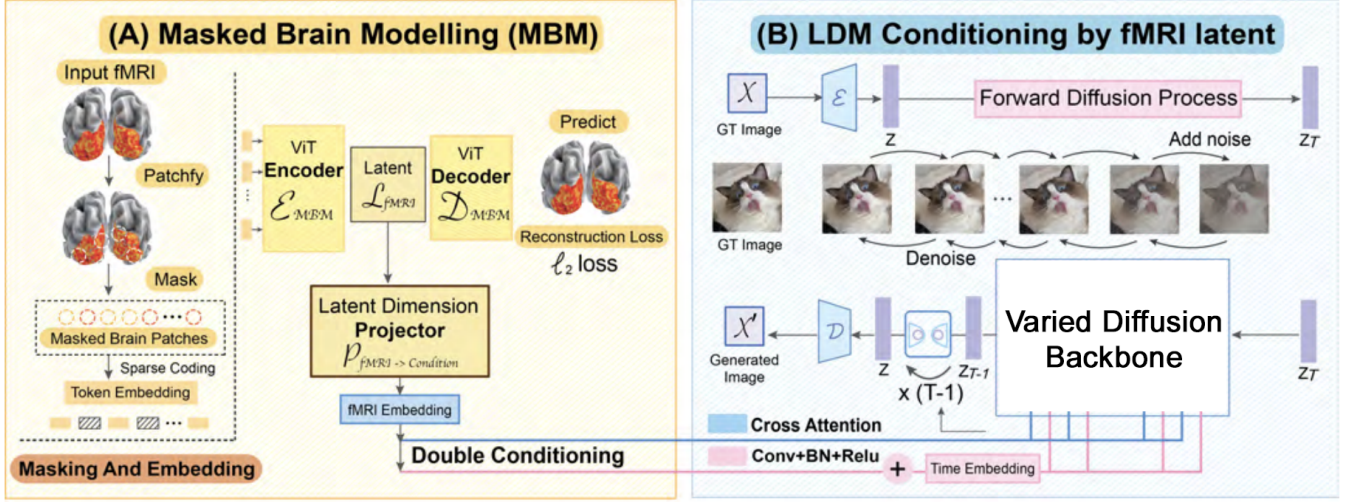


Figure 1: **fMRI Image Reconstruction Diffusion Architecture.** We replace the U-Net with alternative backbones to test hypotheses about what limits current implementations.

is a mapping of human fMRI scans with a corresponding 5,000 images incorporating scenes from well-known datasets such as SUN, COCO, and ImageNet.[12] We use this dataset as our primary validation set. A more recent development from Meta Labs in March of this year was their work in leveraging a dataset of over 1,000 hours of fMRI across 720 subjects to create a tri-modal (video, audio and language) foundation model capable of predicting human brain activity in a variety of naturalistic and experimental conditions, which shows the increasing interest and impact of this field.[7]

Diffusion Models: A diffusion probabilistic model is a parameterized Markov chain trained using variational inference to produce samples matching the data after finite time.[13] Denoising diffusion implicit models (DDIMs), are a more efficient class of iterative implicit probabilistic models designed to produce high quality samples in fewer steps.[17] We utilize the DDIM and run an ablation study on the number of DDIM steps in this work. Historically diffusion models have utilized a U-Net backbone to approximate noise. The U-Net is a convolutional network that was initially created for the biomedical image segmentation space.[14] In this paper we experiment with alternative backbones for our diffusion model. An alternative was proposed by Peebles & Xie (2023)[8] which proposed the DiT which replaces the U-Net in a diffusion model with the transformer architecture. Their approach is found to be compute-efficient and outperforms all prior U-Net-based diffusion models. There are multiple options for the implementation of the DiT block such as cross-attention, in-context conditioning, Adaptive layer norm (adaLN), and adaLN with zero initialization. Peebles & Xie found adaLN-zero to have the strongest performance, hence we proceed with the adaLN-zero DiT block. The U-ViT by Bao et al (2023)[9] proposed an alternative diffusion model backbone which utilizes transformer blocks alongside long skip connections between shallow and deep layers. Their approach achieves comparable or better FID scores than

U-Net-based diffusion models on standard benchmarks. This architecture was also found generalizes well across unconditional generation, class-conditional generation, and text-to-image settings. Additionally, PixArt- α was designed by Yu et al (2023)[10] and proposed an alternative diffusion model backbone that builds on the DiT architecture with additional efficiency-based design choices. Their approach achieves competitive image quality to SoTA text to image models while significantly reducing training cost and improving training efficiency. To bring these models to scale they are often pretrained on ImageNet, a set of 3.2 million images with an emphasis on scale and diversity.[16] We’ve standardized all pretrained models used to be pretrained on ImageNet. Latent Diffusion Models (LDMs) operate in a lower dimensional space utilizing autoencoders to maintain high performance while reducing compute requirements at train and test time.[15] We utilize this technique, as do Chen et al, to reduce our compute requirements.

4 Methods

We base our approach on the current SoT pipeline by Chen. et al. We replaced the U-Net in their LDM with DiT, U-ViT, and PixArt- α architectures each pretrained on ImageNet1K (as was the original LDM). We choose these alternative backbones to evaluate the following hypotheses:

- **DiT:** The 2023 paper from Peebles & Xie found that the DiT architecture consistently outperforms U-Net based diffusion models with respect to FID. We investigate whether this behavior continues when the input to the DDPM comes from fMRI signals. In order to evaluate this hypothesis we compared the FID on the validation set between the U-Net based model vs the DiT based model.
- **U-ViT:** The U-ViT utilizes a transformer architecture with added skip-connections. This will answer the question of

whether skip connections improve performance for fMRI image reconstruction. We tested this by comparing the performance on 50-way top-1 accuracy, SSIM, and PSM between models with (U-Net, U-ViT) vs without (DiT, PixArt- α) skip connections. We choose these three metrics for this evaluation as each is a different measure of reconstruction quality.

- **PixArt- α** was chosen due to its performance in creating realistic images. We explored whether realism is a bottleneck in fMRI image reconstruction by comparing performance on all 4 metrics.

Additionally, for each architecture we conducted an ablation study on crop ratio and DDIM steps which we explain in more detail in the experiments section.

Our primary contribution is an investigation of how the structures of different backbone architectures affect image reconstruction from fMRI signals. Our work aims to identify which backbone architectures are most compatible with fMRI decoding and to provide insight into whether current bottlenecks arise from model design or the limitations of fMRI signals.

For our baseline method we replicated the Chen. et al model with their U-Net backbone and replicated their 50-way top-1 accuracy metric on the BOLD5000 dataset and GOD validation holdout. Additionally we benchmarked the U-Net based model on FID, SSIM, and PSM.

5 Experiments

5.1 DDIM Steps Ablation Study

Arch	DDIM Steps	Acc (%)	FID	SSIM	PSM
DiT	150	17.8%	3.58	0.552	0.725
DiT	250	16.3%	3.81	0.553	0.731
U-ViT	150	16.4%	3.96	0.547	0.680
U-ViT	250	4.5%	5.87	0.499	0.495
PixArt	150	10.4%	4.63	0.557	0.738
PixArt	250	16.0%	3.74	0.582	0.770

Table 1: **DDIM Steps Ablation on GOD.** DiT performs best at 150 steps, PixArt improves with more steps, and U-ViT shows instability at higher step counts.

DDIM steps refers to the number of steps the DDIM model takes in generating samples. A higher number of steps is generally correlated with higher sample quality but lower efficiency. The Chen et al paper used 250 steps but the DDIM paper by Song et al found that performance plateaus after 100 steps. Therefore we ablate with the values of 150, 250. The purpose of this study is to test how the number of DDIM steps affects different architectures since an architecture performing strongly with fewer steps would enable more efficient training and sampling.

5.2 Crop Ratio Ablation Study

Arch	Crop Ratio	Acc (%)	FID	SSIM	PSM
DiT	0.2	16.3%	3.81	0.553	0.731
DiT	0.3	15.1%	4.07	0.551	0.723
U-ViT	0.2	4.5%	5.87	0.499	0.495
U-ViT	0.3	20.8%	3.21	0.531	0.677
PixArt	0.2	16.0%	3.74	0.582	0.770
PixArt	0.3	12.9%	4.28	0.561	0.751

Table 2: **Crop Ratio Ablation on GOD.** U-ViT benefits significantly from a higher crop ratio (0.3), while DiT and PixArt show better performance at lower crop ratios.

We choose to do an ablation study on the crop ratio because the Chen et al paper reports that the crop ratio caused significant variance in 50-way-top-1 accuracy, varying it by 14.9 to 23.9 percent. The chosen values for ablation (0.2, 0.3) are the ones which produced the highest accuracy in Chen et al.

5.3 Results

Arch	Acc (%)	FID	SSIM	PSM
U-Net	29.7%	1.45	0.524	0.732
DiT	21.8%	2.38	0.531	0.730
U-ViT	18.2 %	3.04	0.525	0.681
PixArt	23.5%	2.18	0.540	0.739

Table 3: **Evaluation on BOLD5000.** U-Net outperforms the alternatives in accuracy and FID while PixArt achieves the strongest performance in SSIM and PSM.

Arch	Acc (%)	FID	SSIM	PSM
U-Net	27.2	2.22	0.526	0.735
DiT	17.8	3.58	0.552	0.725
U-ViT	20.8	3.21	0.531	0.677
PixArt	16.0	3.74	0.582	0.770

Table 4: **Evaluation on GOD.** U-Net outperforms the alternatives in accuracy and FID while PixArt achieves the strongest performance in SSIM and PSM.

5.3.1 Architecture Comparison

The BOLD5000 results confirm that U-Net outperforms all transformer backbones on classification accuracy, achieving 29.7% compared to PixArt- α at 23.5%, DiT at 21.8%, and U-ViT at 18.2%. On FID, a similar ordering holds: U-Net (1.45), PixArt- α (2.18), DiT (2.38), U-ViT (3.04). However, the structural and perceptual metrics tell a different story. PixArt- α achieves the highest SSIM (0.540) and PSM (0.739), exceeding U-Net on both. DiT similarly outperforms U-Net on SSIM (0.531 vs 0.524) with comparable PSM (0.730 vs

0.732). This confirms a consistent divergence between semantic metrics (accuracy, FID) and structural/perceptual metrics (SSIM, PSM) across datasets.

Notably, the ranking among transformer backbones shifts between GOD and BOLD5000. On GOD, U-ViT achieves the best transformer accuracy (20.8%) while PixArt- α trails at 16.0%. On BOLD5000, this reverses: PixArt- α leads the transformers at 23.5% while U-ViT falls to 18.2%. This suggests that the relative strengths of each backbone are dataset-dependent, and that PixArt- α 's efficiency-oriented design may generalize better to the larger and more diverse BOLD5000 dataset.

Despite PixArt- α 's strong showing on BOLD5000, the core finding holds across both datasets: transformer backbones that excel on perceptual quality still trail U-Net on semantic preservation. PixArt- α achieves the best SSIM and PSM on both GOD and BOLD5000, yet its accuracy gap to U-Net remains 6.2 percentage points on BOLD5000 and over 11 on GOD. This indicates that realism is not the primary bottleneck in fMRI image reconstruction; the challenge remains preserving class-discriminative semantic content from the weak fMRI conditioning signal.

5.3.2 Role of Skip Connections

U-ViT, which combines a transformer backbone with long skip connections between shallow and deep layers, provides mixed evidence for the role of skip connections in semantic preservation. On GOD, its best configuration (crop ratio 0.3) achieves 20.8% accuracy, the highest among all transformer backbones and substantially closer to the U-Net baseline than either DiT or PixArt- α . However, on BOLD5000, U-ViT falls to 18.2%, below DiT (21.8%) and PixArt- α (23.5%). This suggests that skip connections can aid semantic preservation under favorable conditions but do not provide a consistent advantage across datasets.

Across both datasets, U-ViT's perceptual quality remains the weakest among transformer backbones: its PSM of 0.677 on GOD and 0.681 on BOLD5000 are well below DiT and PixArt- α . Skip connections appear to constrain the model's ability to produce perceptually clean outputs, reinforcing the semantic-structural tradeoff observed across all architectures.

U-ViT also exhibits substantial instability. At 250 DDIM steps with crop ratio 0.2, accuracy collapses to 4.5% (near chance on 50-way classification) and PSM falls to 0.495. Notably, the training loss for this configuration decreased normally, but validation accuracy never improved, a signature of overfitting to the denoising objective without learning generalizable representations. We hypothesize that 250 denoising steps gives U-ViT too many opportunities to drift from the fMRI conditioning signal, with the skip connections faithfully propagating features through a chain that has already lost semantic content. This instability is itself a finding: skip connections in the fMRI setting may amplify sensitivity to hyperparameter choices, producing the strongest transformer results under favorable configurations but collapsing entirely under

unfavorable ones.

5.3.3 Ablation Studies

The DDIM steps and crop ratio ablations reveal that hyperparameter sensitivity is strongly architecture-dependent. For DDIM steps, DiT performs slightly better with fewer steps (17.8% accuracy at 150 vs 16.3% at 250), consistent with Song et al.'s finding that performance plateaus after approximately 100 steps. PixArt- α shows the opposite pattern: 250 steps substantially outperforms 150 across all metrics (16.0% vs 10.4% accuracy, 0.582 vs 0.557 SSIM, 0.770 vs 0.738 PSM), suggesting that PixArt- α genuinely benefits from the additional denoising iterations to produce perceptually refined outputs. U-ViT's behavior is dominated by the collapse at 250 steps discussed above; at 150 steps it recovers to 16.4% accuracy, indicating that the shorter denoising chain keeps outputs closer to the conditioning signal.

Crop ratio shows a similarly architecture-dependent pattern. DiT and PixArt- α both perform slightly worse at crop ratio 0.3 compared to 0.2 on accuracy, while U-ViT benefits dramatically, jumping from 4.5% to 20.8%. For U-ViT, the tighter crop at 0.3 likely acts as a regularizer, forcing the model to focus on central semantic content and preventing the drift observed at the default crop ratio. DiT, by contrast, is the most stable architecture across both ablations: its accuracy ranges only from 15.1% to 17.8% and its perceptual metrics barely move, making it a reliable if unspectacular choice. These results indicate that the denoising schedule and preprocessing pipeline cannot be treated as architecture-agnostic in the fMRI setting; each backbone requires its own tuning, and the interaction between architectural design and hyperparameter choice is a meaningful factor in reconstruction quality.

6 Code Overview

Backbones

In `config.py` line 117 and `stageB_ldm_finetune.py` lines 198-203 we added a config system that allows the user to specify which backbone (U-Net, DiT, U-ViT, PixArt- α) to insert into the model. We made this possible in `ldm_for_fmri.py` lines 46-416 which reads the config to load in the appropriate pretrained backbone. In `dit.py`, `pixart.py`, and `uvit.py` we've merged together the appropriate code for each backbone into our repository. Additionally for each pretrain we wrote a corresponding config: `config_dit_adaln_zero.yaml`, `config_pixart_adaln_...yaml`, and `config_uvit.yaml`.

Experiment Scripts

We wrote two shell scripts to execute all of our ablation studies split across batches: `run_ablations_batch1.sh` and `run_ablations_batch2.sh` which calls our train and evaluation programs and spawns a corresponding WandB run. Additionally we wrote `convert_vae_diffusers_to_ldm.py` which

streamlines converts a diffusers-format state_dict so it can be loaded by AutoencoderKL in our pipeline for the DiT. Lastly we wrote scripts to standardize where the pre-trains are downloaded from: download_dit_weights.sh, download_pixart_weights.sh, and download_uvit_weights.sh.

Evaluation

The original evaluation script did not include FID so we added FID as a metric in ddpm.py lines 507-516 and gen_eval.py lines 67-74 by adding it to the metrics list.

7 Timeline, Human Work, and AI Use

Timeline

Task	Time	Primarily Human?
Reading papers	12 hrs	Yes
Reading docs	6 hrs	Yes
Reading existing code	10 hrs	Yes
Environment setup	8 hrs	Yes
Writing new code	15 hrs	collab w/Claude
Running experiments	30 hrs	
Compiling results	8 hrs	Yes
Writing reports	20 hrs	Yes

Human Work

All dataset setup, preprocessing, and baseline replication were completed entirely through human work. This includes reproducing the original pipeline, verifying correctness of the training procedure, and ensuring alignment with prior reported results.

For all models, the architectural design, experimental setup, and integration plan were developed manually and we focused on understanding how the backbone interacts with the latent diffusion pipeline. Additionally, report writing and interpretation of results was a primarily human task.

AI-Assisted Work and Verification

We use generative AI (Claude Opus 4.6) to assist with integrating new backbones into the LDM. AI assistance is limited to generating boilerplate and framework specific code but not any architectural design or experimental decisions.

To ensure correctness and trustworthiness, we manually reviewed every line to verify alignment with the intended architecture and design. We validated the code by checking consistency with the reference implementation and expected training behavior.

8 Research Log

From Monkeys to Humans. Our project title references “Monkey See, Monkey Diffuse,” which originally reflected

our intention to work with non-human primate fMRI data. Early in the project we explored using monkey fMRI datasets but encountered significant barriers: the data formats were incompatible with the mind-vis pipeline, the dataset sizes were substantially smaller, and most critically, the fMRI encoder in the Chen et al. pipeline would have required retraining from scratch on monkey neural data. Given our compute budget, this was infeasible. We pivoted to working entirely with human fMRI datasets (GOD, BOLD5000), which allowed us to leverage the pretrained encoder and focus our compute on the backbone experiments that were central to our hypotheses. The title survived the pivot.

VAE Compatibility and Pretrained Weights. The most time-consuming engineering challenge was integrating the transformer backbones into the mind-vis pipeline. The original codebase was built around a U-Net that interfaces with a specific VAE architecture. The DiT, U-ViT, and PixArt- α backbones each expect a Stable Diffusion VAE, which is incompatible with the original pipeline’s autoencoder. We wrote a conversion script (convert_vae_diffusers_to_ldm.py) to bridge this gap, translating diffusers-format state dictionaries into the format expected by AutoencoderKL in our pipeline. Beyond the VAE, each backbone came with its own pretrained weight format and naming conventions, requiring substantial effort to load weights correctly. We standardized all three backbones to use ImageNet1K-pretrained weights to ensure a fair comparison, but finding valid pretrained checkpoints and verifying that they loaded without silent failures consumed a significant portion of our implementation time.

Metric Expansion. Our proposal specified only 50-way top-1 accuracy and FID as evaluation metrics. After running our first DiT experiment on BOLD5000, we observed that DiT scored worse on accuracy but the reconstructed images did not look categorically worse in a structural sense. The mind-vis codebase already computed SSIM and PSM (from LPIPS) internally, and these metrics captured a dimension of reconstruction quality that accuracy and FID missed. Since FID is computed from Inception features and correlates strongly with classification accuracy, we found that accuracy and FID were largely redundant as signals. Adding SSIM and PSM gave us an independent axis measuring structural and perceptual similarity, which proved essential to our main finding: the dissociation between semantic fidelity and structural quality across architectures.

Hypotheses Reversed. Our proposal predicted that transformer backbones would outperform the U-Net, with an expected ranking of U-ViT $\geq$ DiT $>$ PixArt- α $>$ U-Net. The DiT results on BOLD5000 were the first indication that this was wrong: the U-Net outperformed DiT on accuracy by nearly 8 percentage points. At that point we suspected that the U-Net’s convolutional inductive biases and skip connections were better suited to the noisy fMRI setting than we had anticipated. The GOD ablation results confirmed this across all three transformer backbones. Rather than being a negative result, this reversal became the central finding of the paper: transformer backbones excel at structural and perceptual re-

construction but fail to preserve class-discriminative semantic content, revealing a fundamental tradeoff that was not visible from accuracy alone.

BOLD5000 vs GOD for Ablations. Our proposal called for evaluation on BOLD5000, but we discovered that BOLD5000 contains roughly $4\times$ the labeled data of GOD, making each training run approximately $4\times$ more expensive. A single BOLD5000 run required approximately 200 GPU-hours compared to 50 for GOD. With 12 ablation runs needed (3 architectures $\times$ 2 DDIM steps $\times$ 2 crop ratios, minus shared configurations), running all ablations on BOLD5000 would have required way more compute than we had. We therefore ran all ablation studies on GOD and reserved BOLD5000 for the final cross-architecture comparison.

U-ViT Collapse and Debugging. The U-ViT run at 250 DDIM steps with crop ratio 0.2 produced a near-chance accuracy of 4.5%, which initially appeared to be a bug. We spent time verifying the configuration, checking that weights loaded correctly, and comparing the training pipeline to our other runs. The training loss curve decreased normally and appeared stable, which ruled out obvious implementation errors. After examining the validation accuracy curve, which remained flat throughout training, we concluded that the model was overfitting to the denoising objective without learning transferable representations. We interpret this as U-ViT settling into a local minimum where it minimizes the training loss by generating plausible-looking images that are semantically unrelated to the fMRI conditioning signal. Rather than discarding this result, we report it as evidence of U-ViT’s instability in the fMRI setting, which is itself a meaningful finding about the interaction between skip connections and denoising step count.

9 Conclusion

In this project, we evaluated 3 unique diffusion backbone architectures and found that transformer-based models do not outperform the U-Net baseline on semantic classification tasks (classification accuracy & FID). However, transformer-based models do match/exceed U-Net on perceptual/structural metrics (SSIM, PSM), which indicates a potential tradeoff between structural quality and semantic quality. This suggests that the primary bottleneck in fMRI reconstruction is not image realism, but the preservation of semantic information from weak conditioning signals.

Our ablations further demonstrate that performance is highly sensitive to architecture-specific hyperparameters, with DDIM steps and crop ratio affecting each backbone differently. Overall, our results show that advances in diffusion architectures do not generalize well to this task, and that convolutional inductive biases found in U-Net remain effective under fMRI signaling. However, PixArt- α ’s strong performance on BOLD5000 (23.5% accuracy, best transformer FID) suggests that some architectural advances may transfer better with larger datasets, though a gap to U-Net remains.

Future work should focus on exploring hybrid architectures

that better balance semantic and perceptual reconstruction, as well as leveraging larger or more diverse datasets for improved fine-tuning of fMRI-conditioned generative models.

10 Thought-Experiment on Compute

Our experiments have required approximately 1258 GPU-hours. This includes 1 GPU-hour for baseline evaluation. For the BOLD5000 dataset, we trained each of the four architectures which required approximately 200 GPU-hours per run, along with approximately 2 GPU-hour for evaluation with 4 runs, for a total of 808 GPU-hours. We also conducted experiments on the GOD dataset. Each run on GOD required approximately 50 GPU-hours (training + evaluation), and with 3 runs per architecture, this results in 450 GPU-hours. In total, this gives:

$$808 + 450 = 1258 \text{ GPU-hours.}$$

To estimate dollar cost, we use AWS G5 pricing for `g5.xlarge` at \$1.00/hour which uses a single A10G. Under this assumption, our total compute usage corresponds to:

$$1258 \times 1.00 = \$1258.00.$$

Thus, our project so far is roughly equivalent to \$1258 of AWS GPU compute. With an additional \$450 in AWS credits (450 GPU-hours), we would allocate this toward further ablations on the BOLD5000 dataset, specifically varying DDIM steps and crop ratio. This would allow for approximately 2–3 additional BOLD5000 runs which would help improve the completeness of our experiments.

References

- [1] Chen et al: *Seeing Beyond the Brain: Conditional Diffusion Model with Sparse Masked Modeling for Vision Decoding*, 2023.
- [2] Ozcelik et al: *Reconstruction of Perceived Images from fMRI Patterns and Semantic Brain Exploration using Instance-Conditioned GANs*, 2022.
- [3] <https://github.com/zjc062/mind-vis>. Github, mind-vis
- [4] Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., and Hochreiter, S. Gans trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in neural information processing systems*, pp. 6626–6637, 2017.
- [5] Wang Z, Bovik A, Sheikh H, Simoncelli E (2004) Image quality assessment: from error visibility to structural similarity. *IEEE Trans Image Process* 13(4):600–612
- [6] Zhang R, Isola P, Efros AA, Shechtman E, Wang O (2018) The unreasonable effectiveness of deep features

- as a perceptual metric. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 586–595
- [7] d’Ascoli et al: *A foundation model of vision, audition, and language for in-silico neuroscience*, 2026.
 - [8] Peebles & Xie: *Scalable Diffusion Models with Transformers*, 2023.
 - [9] Bao et al: *All are Worth Words: A ViT Backbone for Diffusion Models*, 2023.
 - [10] Yu et al: *PixArt- α : Fast Training of Diffusion Transformer for Photorealistic Text-to-Image Synthesis*, 2023.
 - [11] Roman Beliy, Guy Gaziv, Assaf Hoogi, Francesca Strappini, Tal Golan, and Michal Irani. From voxels to pixels and back: Self-supervision in natural-image reconstruction from fmri. *Advances in Neural Information Processing Systems*, 32, 2019
 - [12] Nadine Chang, John A Pyles, Austin Marcus, Abhinav Gupta, Michael J Tarr, and Elissa M Aminoff. Bold5000, a public fmri dataset while viewing 5000 visual images. *Scientific data*, 6(1):1–18, 2019
 - [13] Jonathan Ho, Ajay Jain, Pieter Abbeel. Denoising Diffusion Probabilistic Models. 16 Dec 2020.
 - [14] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-Net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical Image Computing and Computer-Assisted Intervention*, pages 234–241. Springer, 2015.
 - [15] Robin Rombach¹, Andreas Blattmann¹, Dominik Lorenz¹ Patrick, Esser Bjorn Ommer. High-Resolution Image Synthesis with Latent Diffusion Models. 13 Apr 2022.
 - [16] Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *Proc. CVPR’09*, pages 248–255. Ieee, 2009.
 - [17] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” *arXiv preprint arXiv:2010.02502*, 2020